



www.efy.in

**EFYGROUP**  
YOURS SINCE 1969

“

Branding is what people say about you when you are not in the room.

”



-Jeff Bezos



**Continuous monitoring.** The device remains in monitoring mode until it is turned off or reset by the user. It periodically checks for falls or temperature changes while awaiting further input. This integrated approach ensures that individuals can quickly alert their contacts in emergencies while providing critical location and health information for timely assistance.

**Real-time tracking.** If integrated with periodic updates, the system can continue sending location updates to

contacts, providing real-time tracking of the user's movements.

**Reset.** Once the alert has been sent and the situation is handled, the system resets for subsequent use. The push button can be pressed again to initiate another alert, if needed.

### Outputs of each function

#### Location data - GPS coordinates.

The system captures and outputs the device's current GPS coordinates (latitude and longitude), for example, Latitude: 17.52115 and Longitude: 78.63043

#### Status update.

**Key button press acknowledgement.** The system outputs a confirmation message indicating that the key button has been pressed, for example: 'ALERT, EMERGENCY AT LOCATION.'

**High-temperature acknowledgement.** The system outputs a confirmation message indicating a high-temperature alert, for example,

|    |          |        |   |
|----|----------|--------|---|
| 7  | FORCE_ON | 1PPS   | 6 |
| 8  | AADET_N  | V_BCKP | 5 |
| 9  | NC       | VCC    | 4 |
| 10 | RESET    | GND    | 3 |
| 11 | EX_ANT   | TXD1   | 2 |
| 12 | GND      | RXD1   | 1 |

**L86**  
(Top View)

Fig. 5: L86 GPS pinouts (Source: L86 datasheet)

#### SOS\_device.ino

```
1 #include <LSM303AGR_ACC_Sensor.h>
2 #include <LSM303AGR_MAG_Sensor.h>
3 #include <Wire.h>
4
5 // Define pins for hardware serial connections
6 #define GSM_TX_PIN 37 // TX pin for GSM module
7 #define GSM_RX_PIN 38 // RX pin for GSM module
8 #define GPS_TX_PIN 43 // TX pin for GPS module
9 #define GPS_RX_PIN 44 // RX pin for GPS module
10
11 // Define other pins
12 #define BUZZER_PIN 36 // Buzzer pin
13 #define SOS_BUTTON_PIN 7 // SOS button pin
14
15 // Define thresholds
16 #define FALL_THRESHOLD 15.0 // Threshold for detecting a fall
17 #define TEMP_THRESHOLD 38.0 // Threshold for high temperature (Celsius)
18
19 // Create objects
20 LSM303AGR_ACC_Sensor Acc(&Wire);
21 LSM303AGR_MAG_Sensor Mag(&Wire);
22
23 // Use hardware serial
24 HardwareSerial GSM(1); // HardwareSerial 1 for GSM
25 HardwareSerial GPS(2); // HardwareSerial 2 for GPS
26
27 float acceleration_x, acceleration_y, acceleration_z;
28 float temperature;
```

Fig. 6: Snippet of source code